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Problem Understanding & System Definition

1. Problem Understanding

The project aims to develop machine learning models for two main battery-management tasks:

1. Estimation of Battery State of Charge (SOC)
2. Prediction of Remaining Charging Time (RCT)

SOC represents the estimated amount of charge remaining in the battery, while RCT represents the amount of charging time still required to reach the defined charging endpoint.

Accurate SOC and RCT prediction can help in battery monitoring, charging planning, energy management, and providing useful information to EV users.

2. Proposed Problem Statement

Using historical battery charging data such as time, voltage, current, power, temperature, charging status, and session or cycle information, the project will develop machine learning models

Different machine learning models will be tested and compared, starting with simple baseline models and then using tree-based, kernel-based, and boosting methods. A neural-network model may also be considered if the dataset is large enough and its time-based structure makes it suitable

3. Inputs and Targets

Input Features

· Time

· Battery voltage

· Charging current

· Charging power

· Battery temperature

· Charging status / charging phase

· Other supplied battery operating variables

Target Variables

· Target 1: State of Charge (SOC)

· Target 2: Remaining Charging Time (RCT)

4. Important Definitions

SOC: The percentage or normalized value that represents the current charge level of the battery.

Remaining Charging Time: Time remaining from the current observation until the defined charging endpoint. The charging endpoint needs to be identified from the dataset, such as a target SOC or the end of a charging session

5. Proposed End-to-End Pipeline

The pipeline is as follows:-

→ Raw supplied dataset

→ Raw-data preservation

→ Initial inspection

→ Data-quality checks
→ Preprocessing

→ Feature engineering

→ Leakage-safe train/validation/test split

→ Baseline models

→ Candidate ML models

→ Hyper parameter tuning

→ Evaluation

→ Error analysis

→ Comparison

→ Final model selection

→ Documentation and reproducibility

6. Success Criteria

The project will be considered successful if:

- The raw dataset is preserved without modification.
- Data-quality problems are identified and documented.
- No target leakage occurs between training and evaluation.
- SOC and RCT are evaluated separately.
- MAE and RMSE are reported for both targets.
- R^2 is reported where appropriate.
- Multiple candidate models are compared using the same evaluation protocol.
- The final report clearly states limitations and assumptions.

7. Technical Questions / Assumptions

1. What type of battery is represented in the dataset?

2. What are the units of voltage, current, power and

temperature? 3. Is SOC already provided in the dataset?

4. Is Remaining Charging Time already provided?

5. If RCT is not provided, can it be calculated from the timestamps? 6.

Does the dataset contain multiple charging sessions or battery cycles?

7. Are there missing or invalid measurements?

8. Are there duplicate observations?

9. Which train-test splitting method would be most suitable for this

dataset? 10. Can the same charging session appear in both training and

testing data?